

Euclid’s Elements — Book I

Proposition 10

Formal Reconstruction — Technical Documentation

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Bisecting a finite straight line

1 Introduction

Proposition I.10 constructs the midpoint of a given finite straight line. In Euclid’s development the construction is built from three earlier propositions: I.1 supplies an equilateral triangle, I.9 bisects its apex angle, and I.4 applies SAS to show that the two pieces of the original segment are congruent.

The current Lean endpoint has a very different surface shape. The theorem `euclid_proposition_10` is a thin wrapper around `HilbertMidpointExists`. A diagrammatic audit, however, must not stop at that wrapper. The interface theorem itself is only a packaging layer for Hilbert’s neutral midpoint theorem `hilbert_midpoint_exists` in `HilbertAxioms`. That lower theorem contains a substantial synthetic construction involving angle construction, segment layoff, a neutral alternate-angle criterion, plane separation, Pasch, vertical angles, and AAS.

Thus Proposition I.10 is a useful example of three distinct levels that should not be conflated:

1. Euclid’s historical construction;
2. the short public Book I wrapper;
3. the hidden mathematical engine already proved in the Hilbert layer.

2 The Classical Configuration

Let AB be the given finite straight line. The objective is to construct a point on the open segment AB dividing it into two congruent parts.

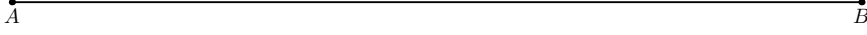


Figure 1: The given finite straight line AB .

2.1 Construction

Construct the equilateral triangle ABC on the segment AB by Proposition I.1. Bisect the angle $\angle ACB$ by Proposition I.9, and let the bisecting ray meet AB at D .

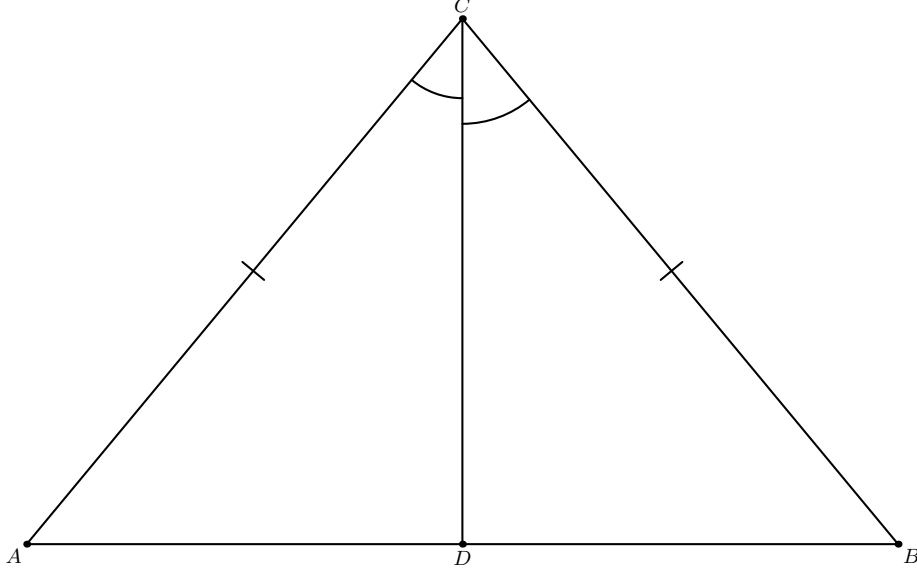


Figure 2: Euclid's construction for Proposition I.10. The equilateral triangle gives $AC \cong BC$; the ray CD bisects the angle at C . SAS for $\triangle ACD$ and $\triangle BCD$ yields $AD \cong DB$.

The segment CD is common to the two triangles. Hence

$$AC \cong BC, \quad CD \cong CD, \quad \angle ACD \cong \angle DCB.$$

By Proposition I.4,

$$AD \cong DB.$$

Since D lies between A and B , it is the required midpoint.

3 Statement

Theorem 1 (Euclid I.10). *Let AB be a nondegenerate segment, so that $A \neq B$. Then there exists a point M such that*

$$A - M - B \quad \text{and} \quad AM \cong MB.$$

Equivalently, every nondegenerate segment admits a strict Hilbert midpoint.

In the project this conclusion is represented by `HilbertIsMidpoint`. Thus the formal statement records strict betweenness as part of the midpoint data; equality of two segment magnitudes alone would not be sufficient.

Diagrammatic audit. Three geometric figures are used in the audited chapter. Figure 1 records the input segment. Figure 2 contains the complete Euclidean construction. Figure 3 records the lower Hilbert theorem hidden by the public wrapper. This additional formal figure is necessary because the actual mathematical proof of midpoint existence is not the one-line wrapper: it is already stored one layer below in `hilbert_midpoint_exists`.

4 Classical Proof

Euclid’s proof is a compact synthesis of Propositions I.1, I.9, and I.4. Construct the equilateral triangle ABC on AB . Bisect $\angle ACB$ and let the bisector meet AB at D . Then

$$AC \cong BC, \quad CD \cong CD, \quad \angle ACD \cong \angle DCB.$$

SAS gives congruence of $\triangle ACD$ and $\triangle BCD$, hence $AD \cong DB$. The construction of D on the segment supplies the required order relation, so D is the midpoint of AB .

The classical dependency chain is therefore

$$\text{I.1} \longrightarrow \text{I.9} \longrightarrow \text{I.4} \longrightarrow \text{I.10}.$$

5 Proof Architecture of the Reconstruction

The current production theorem is

```
theorem euclid_proposition_10
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B : Geo.Point)
  (hAB : A != B) :
  exists M : Geo.Point,
    HilbertIsMidpoint Geo M A B := by
exact
  HilbertMidpointExists
    Geo A B hAB
```

At this level there is no proposition-local helper and no case split. It is important, however, not to mistake this short endpoint for the complete proof architecture. The current `HilbertInterface` explicitly documents `HilbertMidpointExists` as a stable wrapper for the neutral theorem `hilbert_midpoint_exists` from `HilbertAxioms`.

Thus the public path is

$$\text{hilbert_midpoint_exists} \longrightarrow \text{HilbertMidpointExists} \longrightarrow \text{euclid_proposition_10}.$$

The middle arrow is packaging: the primitive conclusion

$$\exists E, \quad A - E - B \quad \text{and} \quad AE \cong EB$$

is repackaged as the project predicate `HilbertIsMidpoint`.

6 Hilbert Reconstruction

The hidden mathematical engine is Hilbert’s Theorem 26 as reconstructed in `hilbert_midpoint_exists`. It does not replay Euclid’s equilateral triangle and angle-bisector proof. Instead it constructs two congruent, parallel side segments on opposite sides of the given base and forces their joining segment to cross the base at its midpoint.

6.1 Choose an exterior point and the opposite side

Let `base` be the line through A and B . The proof chooses a point C outside this line. It then extends the segment CA beyond A to a point S :

$$C - A - S.$$

Consequently C and S lie on opposite sides of `base`. The role of S is purely orientational: it selects the side on which the angle at B is to be constructed.

6.2 Copy the angle and lay off a congruent side

Hilbert III.4 constructs a ray at B on the side of AB containing S so that its angle with BA is congruent to $\angle BAC$. The source code first names a representative point D_0 on this ray. Hilbert III.1 then lays off on the same ray a point D satisfying

$$BD \cong AC.$$

Because D remains on the selected side, C and D are on opposite sides of the base.

The representative D_0 is not given a separate diagram panel. Its sole role is to create the ray; after segment construction, all stable geometric data are carried by D and the same-ray relation between D_0 and D .

6.3 Obtain the neutral parallel pair

A point T between A and B is chosen to normalize the two directions of the transversal. Same-ray transport converts the copied angle into a pair of equal alternate angles. The neutral converse theorem for alternate angles then yields

$$AC \parallel BD.$$

This implication uses no Euclidean parallel axiom. The proof of `hilbert_midpoint_exists` is explicitly placed before the declaration of the Hilbert Group IV class and states that only Groups I–III are used.

The point T is another normalization witness rather than a persistent new state of the construction, so it is also omitted from the final diagram. Panel (a) of Figure 3 records the stable result: $AC \cong BD$, $AC \parallel BD$, and C, D on opposite sides of AB .

6.4 Construct the crossing point

Since C and D are on opposite sides of `base`, the definition of `HilbertOppositeSide` supplies a point E with

$$C - E - D$$

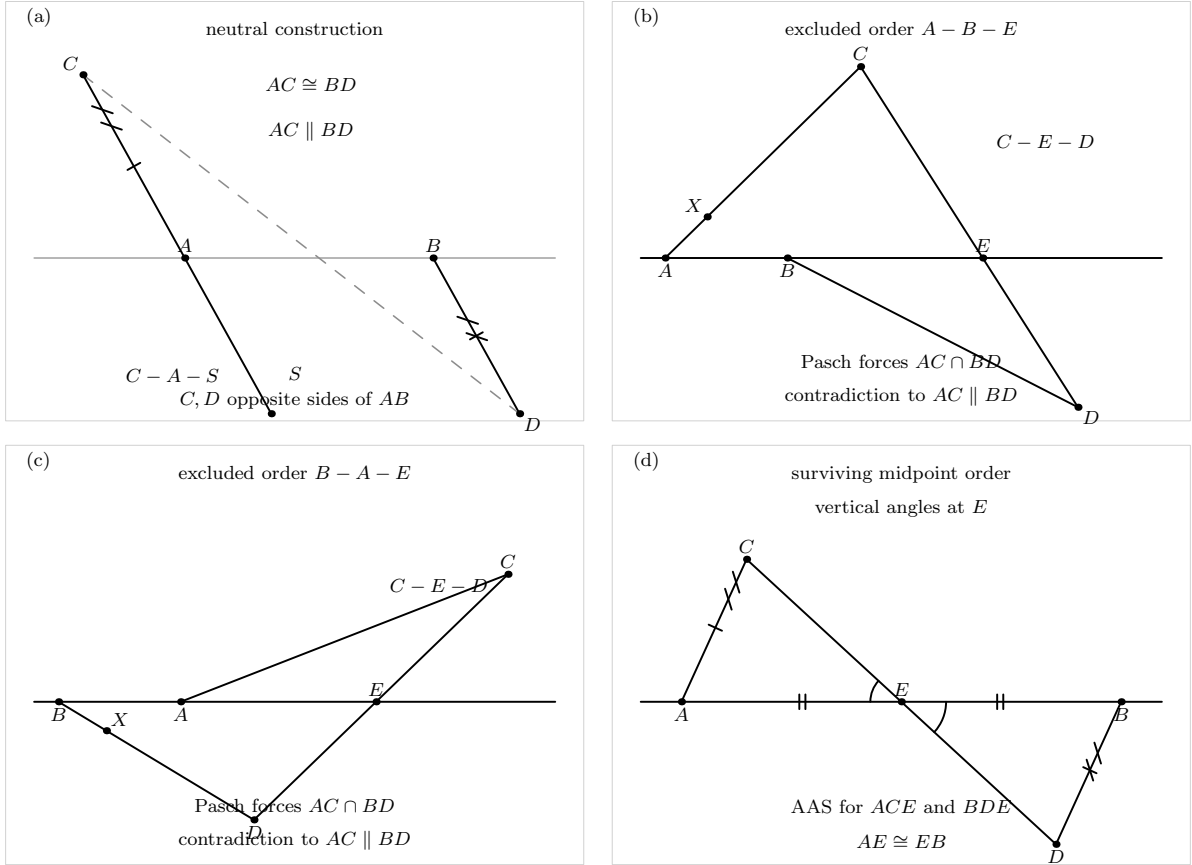


Figure 3: Formal Hilbert geometry underlying Proposition I.10. Panel (a) records the neutral construction of congruent parallel sides on opposite sides of AB . Panels (b) and (c) show the two exterior orders for the crossing point E that are excluded by Pasch together with disjointness of the parallel carriers. Panel (d) is the surviving order $A - E - B$; vertical angles at E and AAS for $\triangle ACE$ and $\triangle BDE$ yield $AE \cong EB$.

and E on **base**. At this stage the proof knows that A, B, E are collinear, but it has not yet established the desired order $A - E - B$.

The parallel lines through AC and BD are represented by their incidence carriers. Their disjointness is used repeatedly below to forbid a second intersection of the transversal CD with either parallel side.

6.5 Exclude the exterior order $A - B - E$

Assume temporarily that

$$A - B - E.$$

The open segment AE then meets the carrier BD at B . Pasch is applied to the triangle AEC . The other candidate side EC cannot meet BD : such an intersection would force a second common point of the crossing line and BD , and hence eventually contradict the disjointness of AC and BD . Pasch therefore forces the remaining side AC to meet BD , directly contradicting the parallelism.

This is the geometric content of panel (b). It is a genuine branch of the proof, not merely a tactic artifact, because it assumes a different strict betweenness order of the actual crossing point E .

6.6 Exclude the exterior order $B - A - E$

The second possible exterior order

$$B - A - E$$

is eliminated symmetrically. Now the segment BE meets the carrier AC at A , and Pasch is applied to the triangle BED . The side ED cannot meet AC a second time, so Pasch would force BD to meet AC , again contradicting parallelism.

Panel (c) records this second genuine order branch.

6.7 The surviving order

The three points A, B, E are distinct and collinear. Hilbert's betweenness trichotomy therefore leaves only three possible strict orders. The first two have just been excluded, so

$$A - E - B.$$

This is exactly the positional half of the midpoint conclusion.

6.8 Vertical angles and AAS

It remains to prove equality of the two halves. Since

$$A - E - B \quad \text{and} \quad C - E - D,$$

the angles at the crossing point are vertical:

$$\angle AEC \cong \angle BED.$$

The copied-angle construction, transported along the rays AE and BE , gives the second angle congruence needed for the triangles ACE and BDE . Together with

$$AC \cong BD,$$

the reusable AAS theorem `hilbert_aas_sides` yields the remaining side congruence. After reversing the second segment representation where needed, the conclusion is

$$AE \cong EB.$$

Combining this with $A - E - B$ gives the required strict midpoint.

7 Lean Formalization

The public theorem itself remains intentionally small. It should be read as an API endpoint, not as a claim that midpoint existence is primitive:

```
exact
  HilbertMidpointExists
    Geo A B hAB
```

The interface theorem in turn packages the lower result. The important architectural distinction is therefore:

- `euclid_proposition_10`: Book I-facing wrapper;
- `HilbertMidpointExists`: stable interface wrapper;
- `hilbert_midpoint_exists`: mathematical proof.

No proposition-local construction occurs in `Proposition10.lean`, but a substantial construction does occur in the imported Hilbert layer. The documentation now records both facts instead of collapsing them into the statement that “no auxiliary construction is required.”

8 Relationship with Earlier Propositions

8.1 Proposition I.1

In Euclid’s proof, I.1 supplies the equilateral triangle on AB . This is a classical dependency only. The current `euclid_proposition_10` does not call `euclid_proposition_1`.

8.2 Proposition I.9

I.9 supplies the classical angle bisector at the apex of the equilateral triangle. Again, it belongs to the Euclidean historical DAG, not the direct Lean call graph of Proposition I.10.

8.3 Proposition I.4

SAS is the decisive final comparison in Euclid’s proof. The lower Hilbert midpoint theorem follows a different route and finishes with the already derived AAS theorem `hilbert_aas_sides` rather than by calling the Book I proposition.

8.4 Transition to Proposition I.11

Midpoint infrastructure is used immediately in the formal development of right-angle existence underlying Proposition I.11. This makes I.10 and I.11 a useful adjacent pair for understanding the project architecture: both have very short public endpoints while their reusable Hilbert-layer existence theorems contain the substantial geometry.

9 Hilbert Components Used

9.1 Strict midpoint representation

The target predicate is

$$\text{HilbertIsMidpoint } Geo\ M\ A\ B,$$

which packages

$$A - M - B \quad \text{and} \quad AM \cong MB.$$

The order component is therefore intrinsic to the result.

9.2 Incidence and order

The construction uses a line through A, B , a point outside that line, segment extension, plane separation, betweenness trichotomy, and uniqueness properties of incidence lines. These operations are not coordinates; they are consequences of Hilbert's incidence and order groups.

9.3 Congruence constructions

Hilbert III.4 copies the angle at A onto the selected side at B , while Hilbert III.1 lays off $BD \cong AC$. Same-ray transport ensures that replacing the initial angle-ray representative by the laid-off point does not change the constructed angle.

9.4 Neutral parallel criterion

The equality of alternate angles is used in the converse direction to prove $AC \parallel BD$. This direction is neutral and does not require Hilbert's Euclidean uniqueness-of-parallels axiom.

9.5 Pasch and plane separation

Pasch is the key tool for excluding the two exterior orders of E . In each branch, one side of a triangle is known to meet one of the parallel carriers; another side is proved not to meet it, so Pasch forces the third side to meet the supposedly disjoint parallel carrier. The contradiction is entirely synthetic.

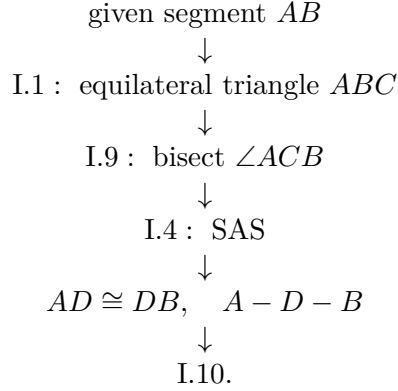
9.6 Vertical angles and AAS

The final side congruence uses the vertical-angle theorem at the intersection E and the reusable AAS theorem for the two triangles. Thus the formal Hilbert proof and Euclid's classical proof

both finish by triangle congruence, but they reach the comparison through different constructions and different congruence criteria.

10 Dependency Summary

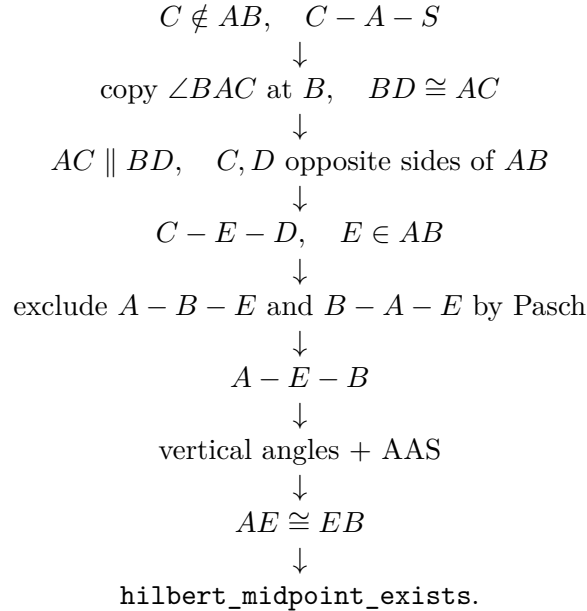
The classical architecture is



The public formal architecture is

$$\text{HilbertMidpointExists} \longrightarrow \text{euclid_proposition_10.}$$

The hidden mathematical architecture is



The final API layering is therefore

$$\text{hilbert_midpoint_exists} \longrightarrow \text{HilbertMidpointExists} \longrightarrow \text{euclid_proposition_10.}$$

No numerical length, coordinate system, continuity assumption, or Hilbert Group IV parallel axiom is introduced in this chain.

11 Diagrammatic Audit

The preserved Book1R chapter already had two useful classical pictures: the input segment and the completed Euclidean construction. Both are retained, now as Asymptote figures so that the chapter follows the same graphics policy as the later audited propositions.

The old documentation stopped the formal diagram analysis at `HilbertMidpointExists`. That was too shallow. The current source states that this interface theorem merely packages `hilbert_midpoint_exists`; the latter contains real geometric states that are not visible in the classical figure.

Figure 3 therefore adds a four-panel formal map. The first panel records the stable neutral construction after the angle-ray and segment-layoff witnesses have been normalized. The next two panels correspond to the two strict exterior betweenness branches actually eliminated in the proof. The final panel records the surviving order and triangle comparison.

No separate panel is added for D_0 , for the internal point T used in angle normalization, or for the final congruence-reversal operation. These steps do not create new persistent geometric configurations. The audited figure set is therefore

3 figures, 6 panels in total.

12 Audit Status

This revision checks the documentation against the current production source `Proposition10.lean`, the current interface declaration `HilbertMidpointExists`, and the underlying theorem `hilbert_midpoint_exists` in `HilbertAxioms`. The production Lean files have not been modified.

The present documentation audit does not claim that a fresh local `lake build` or `#print axioms` command was run. The claim about Groups I–III and the absence of Group IV in the midpoint proof is the explicit source-level statement attached to the current theorem and is also consistent with its position before the Group IV class declaration.

13 Conclusion

Proposition I.10 is simultaneously one of the shortest public Book I theorems and the endpoint of a substantial earlier construction. Euclid reaches the midpoint through an equilateral triangle, an angle bisector, and SAS. The current Hilbert layer reaches it through a different neutral configuration: copy an angle on the opposite side, lay off a congruent side, obtain parallel carriers from alternate angles, force their transversal to meet the base, exclude the two exterior orders by Pasch, and finish with vertical angles and AAS.

The public Lean theorem hides this work deliberately behind a reusable interface. The correct documentation must therefore distinguish theorem size from mathematical dependency depth. Once that distinction is made, I.10 becomes a particularly clear illustration of the library’s layered architecture and prepares the reader for the same phenomenon in Proposition I.11.